DEVISSAGE CRITERION I: A PROPERTY OF COHERENT SHEAVES ON A NOETHERIAN SCHEME THAT IS STABLE UNDER EXTENSIONS AND HOLDS FOR PUSHFORWARDS OF QUASI-COHERENT IDEAL SHEAVES ON INTEGRAL CLOSED SUBSCHEMES HOLDS FOR EVERY COHERENT SHEAF.

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3360-3378 of sources/stacks-coherent/pinned-original-source.tex; proof span is lines 3380-3393, the complete author proof (12 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Pool stacks-coherent backfill wave 33h (tail batch, final screening of ranks 86-118).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. See why-field; every pointer named here occurs verbatim in the retained statement/proof text.

Source reference. Original lemma 01YF: sources/stacks-coherent/pinned-original-source.tex, source lines 3268–3284. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let X be a Noetherian scheme. Let $\mathcal{P}$ be a property of coherent sheaves on X. Assume*

(1) *For any short exact sequence of coherent sheaves*
$$0 \to \mathcal{F}_1 \to \mathcal{F} \to \mathcal{F}_2 \to 0$$
if $\mathcal{F}_i$, $i = 1, 2$ have property $\mathcal{P}$ then so does $\mathcal{F}$.

(2) *For every integral closed subscheme $Z \subset X$ and every quasi-coherent sheaf of ideals $\mathcal{I} \subset \mathcal{O}_Z$ we have $\mathcal{P}$ for $i_*\mathcal{I}$.*

Then property $\mathcal{P}$ holds for every coherent sheaf on X.

Proof. First note that if $\mathcal{F}$ is a coherent sheaf with a filtration
$$0 = \mathcal{F}_0 \subset \mathcal{F}_1 \subset \ldots \subset \mathcal{F}_m = \mathcal{F}$$
by coherent subsheaves such that each of $\mathcal{F}_i/\mathcal{F}_{i-1}$ has property $\mathcal{P}$, then so does $\mathcal{F}$. This follows from the property (1) for $\mathcal{P}$. On the other hand, by Lemma 01YF we can filter any $\mathcal{F}$ with successive subquotients as in (2). Hence the lemma follows. □

Date: stacks-project:coherent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.